

REIMS CHAMPAGNE, France

WMO: 070700

Lat: 49.310N

Lon: 4.050E

Elev: 312

StdP: 14.53

Time Zone: 1.00 (EUC)

Period: 94-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.9	22.5	12.9	10.7	20.7	17.3	13.3	24.2	29.8	49.3	26.6	49.9	4.2	30	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	87.0	68.9	83.4	67.6	79.9	66.5	70.8	83.1	69.2	80.4	67.5	77.3	8.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.6	99.0	75.2	65.2	94.4	73.7	63.8	89.7	72.2	34.8	83.2	33.4	80.4	32.1	77.6	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	20.3	18.0	12.1	93.1	5.1	4.1	8.4	96.1	5.4	98.4	2.5	100.7	-1.2	103.7
			11.4	74.3	5.0	1.5	7.8	75.4	4.8	76.3	2.0	77.2	-1.6	78.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.5	38.0	39.7	44.8	49.7	56.9	61.9	65.9	65.6	59.2	52.6	44.5	38.6
	DBStd	11.88	8.21	7.53	6.19	6.09	5.63	5.73	5.40	5.29	5.24	6.52	6.91	7.90
	HDD50	1537	377	293	181	78	10	0	0	0	1	49	189	359
	HDD65	5131	837	710	626	458	258	127	55	55	187	384	616	818
	CDD50	2092	5	3	20	69	223	358	494	485	276	130	23	6
	CDD65	211	0	0	0	0	5	34	84	75	12	1	0	0
	CDH74	2476	0	0	0	16	109	428	937	844	135	7	0	0
	CDH80	758	0	0	0	0	12	110	306	309	21	0	0	0
Wind	WSAvg	8.3	9.7	9.5	9.1	8.4	7.8	7.3	7.5	7.0	7.4	8.2	8.6	9.5
Precipitation	PrecAvg	26.9	2.2	1.9	1.8	1.8	2.3	2.3	2.6	2.7	2.0	2.4	2.2	2.8
	PrecMax	36.9	5.9	3.9	5.9	5.4	4.7	5.2	6.3	6.1	4.5	4.9	4.1	6.3
	PrecMin	20.7	0.2	0.4	0.3	0.1	0.4	0.7	0.6	0.4	0.4	0.9	0.5	1.3
	PrecStd	3.7	1.1	0.9	1.2	1.2	1.1	1.3	1.3	1.4	1.1	0.9	1.1	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.6	58.4	68.2	76.2	81.6	87.9	91.7	95.1	82.9	73.7	62.5	57.0
		MCWB	53.1	50.8	54.6	61.0	67.7	71.4	69.5	70.2	67.3	64.2	57.5	53.8
	2%	DB	54.3	54.8	63.0	71.4	76.9	82.9	87.2	87.2	78.0	68.6	59.0	54.5
		MCWB	51.5	49.8	53.4	58.0	63.9	68.6	68.6	68.7	65.5	61.2	55.1	51.8
	5%	DB	51.7	52.2	58.9	66.1	73.4	79.0	83.3	82.8	73.5	65.5	56.6	52.5
		MCWB	48.9	48.6	51.7	55.7	62.1	66.9	67.5	67.1	63.6	59.7	53.4	50.1
	10%	DB	49.3	50.2	55.7	62.1	69.6	75.1	79.4	78.4	69.7	62.6	54.3	50.1
		MCWB	46.9	47.3	50.5	53.7	60.1	64.7	66.1	65.6	60.9	58.1	51.7	48.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.0	52.6	56.7	63.4	69.6	72.7	72.6	72.4	69.2	65.1	58.2	54.4
		MCDB	56.2	56.2	63.5	74.1	78.7	83.5	86.1	87.3	78.9	71.2	61.7	56.4
	2%	WB	51.8	50.7	54.7	59.3	65.9	70.0	70.5	70.4	66.6	62.4	55.6	52.3
		MCDB	54.0	53.6	60.4	67.9	74.8	80.4	83.1	83.7	75.6	67.2	58.3	54.3
	5%	WB	49.2	49.2	52.9	56.6	63.0	67.8	68.8	68.6	64.4	60.4	53.8	50.3
		MCDB	51.4	51.8	58.4	64.4	70.6	76.9	80.5	79.6	71.4	64.4	56.0	52.2
	10%	WB	46.8	47.4	50.8	54.3	60.8	65.7	67.1	66.7	62.4	58.5	51.9	48.3
		MCDB	49.0	50.0	55.0	60.8	68.4	73.6	77.1	75.7	68.1	62.0	54.3	50.1
Mean Daily Temperature Range	5% DB	MDBR	9.9	12.4	16.3	18.9	19.1	20.1	20.4	20.1	18.8	15.8	11.0	9.1
		MCDBR	10.3	14.7	23.5	27.3	27.2	27.8	28.1	29.5	25.4	19.0	11.8	9.5
		MCWBR	8.8	10.9	15.1	16.1	15.8	15.4	12.3	13.2	14.0	12.4	9.0	8.1
	5% WB	MCDBR	10.0	12.6	18.7	23.5	23.7	24.7	24.7	25.2	21.3	17.3	10.7	9.1
		MCWBR	9.0	10.0	12.6	14.3	14.8	14.3	11.9	12.0	12.4	11.9	8.6	8.0
Clear-Sky Solar Irradiance	taub	0.320	0.342	0.375	0.404	0.413	0.414	0.412	0.404	0.374	0.362	0.333	0.316	
	taud	2.416	2.369	2.276	2.209	2.221	2.252	2.251	2.300	2.364	2.396	2.414	2.386	
	Ebn at Noon	232	253	262	266	269	269	267	264	260	242	223	214	
	Edh at Noon	21	27	35	42	43	42	41	37	32	26	21	19	
All-Sky Solar Radiation	RadAvg	270	507	886	1366	1609	1787	1697	1458	1122	653	316	219	
	RadStd	38	85	116	166	173	166	157	155	104	54	33	37	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page